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Sampling Approaches for Imbalanced Data Classification Problem in Machine Learning

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Abstract. Real world datasets in many domains like medical, intrusion detection, fraud transactions and bioinformatics are highly imbalanced. In classification problems, imbalanced datasets negatively affect the accuracy of class predictions. This skewness in can be handled either by over-sampling minority class examples or by undersampling majority class. In this work, popular methods of both categories have been evaluated for their capability of improving imbalanced ratio of five highly imbalanced datasets from different application domains. Effect of balancing on classification results has been also investigated. It has been observed that adaptive synthetic oversampling approach can best improve the imbalance ratio as well as classification results. However, undersampling approaches gave better overall performance on all datasets.

Keywords: imbalanced dataset · machine learning · resampling · under sampling · oversampling.

1 Introduction

Real world datasets like medical data of cancerous and non-cancerous cases, credit card fraud data, intrusion detection datasets and sensor based events reporting are inherently imbalanced. In an imbalanced dataset with binary valued classes, one class of data far outnumbers the other class which is generally more important to study. Credit card transactional datasets, for example, have a typical ratio of 10000:1 or more for legitimate transactions to fraudulent ones. The problem of imbalanced datasets in machine learning is an important research issue [1]. A dataset is imbalanced when out of two possible classes of outcomes, occurrence of one is very rare. The performance of chosen classification model depends upon quality and quantity of training dataset. More the volume of training dataset, more is the expected accuracy of classification [2]. In imbalanced datasets with 2- valued classes, accuracy may not show the clear picture of classification results. For example, if a dataset has 99% negative and 1% positive class values then accuracy can reach upto 99% even if the classifier doesn't classify even a single instance of positive class correctly. In many applications like

disease detection and intrusion detection, it is more important to correctly predict the minority class. That is, non-identification of a fraudulent application will be more expensive on part of company than suspecting a normal application to a credible one [3]. The actual assessment can be obtained by analyzing other parameters like balanced accuracy, negative predictive values and positive predictive values.

In machine learning, class imbalance issue can be addressed by either tuning the dataset itself or by changing learning methods of underlying algorithm. At algorithm level, results of classifier are tuned by approaches like adaboosting, bagging and cost-sensitive learning [4]. With respect to data level based imbalance handling, occurrence of both classes is equated algorithmically to improve imbalance ratio. In this paper, we outline and evaluate various data level imbalance handling methods for their capability of improving imbalanced ratio of five highly imbalanced datasets of different application domains.

The paper has been organized in four more sections. Section II describes the Imbalance Data Classification and related problems. In section III, various resampling methods for class imbalance have been studied briefly. Results obtained on various metrics of comparison have been presented and discussed in Section IV. Section V includes the paper conclusion.

2 Resampling Methods For Class Imbalance

Data-level methods dealing with imbalanced datasets aim to obtain balanced distribution by performing some modification or alteration in number of majority and minority class instances [5]. This process to reduce imbalance by changing number of instances of each class type is known as resampling. Resampling can be done in two ways namely undersampling and oversampling.

2.1 Undersampling

When large number of instances are available in the original dataset, then undersampling can be an effective means to improve imbalance. Undersampling does balancing by decreasing the more frequently occurring class. Redundancy in data instances is used to choose samples to be removed and create a balance in classes to train classification models. Here several undersampling techniques implemented in imbalance package of R language have been evaluated [6]. The techniques are summarized as follows:

CNN- Condensed Nearest Neighbor: In nearest neighbour classification, test instances are mapped to the label of instance of the training data to which it is most similar. In kNN (K-Nearest Neighbour) most frequently occurring label among 'k' nearest neighbours of test case is assigned. However, kNN is very slow as for every test instance, it checks all the training instances for similarity. CNN has been designed to speed up kNN by reducing instances to be checked [7]. Outliers are identified and removed for obtaining minimal set of instances that

were correctly classified by 1-NN. These instances become the absorbed points and are used in KNN. In this work, CNN has been used for data undersampling majority data in imbalanced datasets.

TL- Tomek Links: In a dataset, a pair of instances are said to have a tomek link if each of them are nearest neighbour of other while belonging to different class. Such pairs are generally boundary cases or noise and thus likely to be misclassified. Therefore, these instances can be removed without affecting the quality of dataset.

- The Tomek link function searches for the instances that are tomek links using 1-NN for the given dataset.
- To reduce imbalance, majority class examples involved in tomek links are removed[8].

OSS- One-Sided Selection: To solve the problem of imbalanced datasets authors in [9] proposed OSS approach.

- In OSS, synthetic samples are generated by using the approach of Tomek links followed by the CNN method.
- Firstly, any overlapping, borderline and/or noisy instances are removed.
- The links of majority class are discarded until for every instance its nearest neighbour from remaining dataset belongs to its own class[10].
- Then, the instances obtained are processed using CNN.

ENN- Edited Nearest Neighbor: ENN is also based on kNN classification. It extends OSS by considering three nearest neighbours of each instance of frequently occurring class. An instance is removed if its class is different from at least two of its three nearest neighbours [11].

NCL- Neighborhood Cleaning Rule: NCL is similar to ENN as it also looks for inconsistencies in classification of three nearest neighbours. However, NCL has been initially designed for cleaning the dataset based on neighbour similarity property [12]. It works by looking for neighbours of rarer class and removes any opposite class label, if found in this set.

2.2 Oversampling

The process of oversampling is used when the available dataset is imbalanced but small. These methods balance the dataset by creating synthetic samples of minority class.

SMOTE- Synthetic Minority Over-Sampling Technique: It is a powerful sampling method in which the minority class is oversampled by creating synthetic samples [13]. Matching rows according to kNN are randomly chosen and their convex combinations are prepared to obtain new samples. SMOTE has been used for generating artificial fraud transactions for creating balanced credit card datasets [14-15].

MWMOTE- Majority Weighted Minority Oversampling Technique: It is a modification over SMOTE technique and works better when quality of input data is low [16].

- First ranks the minority class samples according to their ease of being distinguishable from other class.
- Similar samples of only those minority class instances that are easily distinguishable are generated by using clustering methods [17].

PDFOS- Probability Density Function Estimation Based Oversampling: PDFOS generates synthetic data by approximating a Gaussian multivariate distribution [16][18]. Probability density function of existing samples of minority class is estimated and extrapolated to generate matching instances. Details about this method can be obtained from [18].

RWO-Random Walk Oversampling: Random walk oversampling generates synthetic samples for minority class while preserving its variance and mean [19]. RWO function is also a part of imbalance package in R [16]. The process involved in RWO oversampling technique are:

- RWO generates synthetic samples by randomly walking through real data and create samples with similar dispersion.
- The weighted information of less frequent samples helps in generation of final synthetic sample by using clustering approach. In order to make the range of generated samples similar to some of the rarer class cluster member.

ADASYN- Adaptive synthetic sampling: ADASYN approaches for imbalanced learning is a two-class imbalanced classification technique. ADASYN approach improves classification learning in two ways:

- Decreasing the bias, if any, brought in by the distribution of classes
- By improving learning by shifting the classification decision boundaries.

ADASYN is an oversampling function available in R's library smote family package [14]. A more detailed description of approach can be seen in [20].

WRACOG- Wrapper for rapidly converging Gibbs algorithm: WRACOG [24] is an enhancement of RACOG algorithm and a wrapper oriented classification technique.

- It first generates a synthetic samples focusing on increasing the minority class by approximating probability distribution.
- This process is repeated until further improvement.
- The samples with highest probability of getting misclassified get selected by WRACOG model.

WRACOG algorithm uses the process of Gibbs algorithm for data sampling while improving many aspects related to usage of Gibbs method of sampling[21].

3 Performance Evaluation

3.1 Datasets Used

We used publicly available, processed real datasets for evaluation:

Dataset-1: CREDIT CARD FRAUD DETECTION DATASET

The dataset has total of 284,807 transactions. It was collected by European cardholders and analyzed during a research collaboration by Andrea Dal Pozzolo and his peers [22].

Dataset-2 : PUBCHEM BIOASSAY DATASET

The dataset has a total of 467,892 record of primary and confirmatory bioassay data from the different types of screening including bioactivity outcomes and potency performed using HTS technology. The main class attributes are labelled as active and inactive compounds for drug like small molecules properties present in bioassay of a patient. The dataset is publicly available and provided by Pubchem [23].

Dataset-3 : YEAST DATASET

It is about applying several bio-statistical tests on yeast bacteria to detect protein. It is available for public usage at [24]. We used yeast dataset with ERL(endoplasmic reticulum lumen) as minority class and combined the other classes as majority class.

Dataset-4 : SATIMAGE DATASET

A fraction of original data of this dataset has been used here [25]. We used damp as minority class instances and others as majority class instances.

Dataset-5 :DIABETES DATASET

It is a dataset of Indian women heritage and records their vital parameters to establish reasons of diabetes by taking samples of affected and non-affected females data [26].

Table 1 represents the composition of all the datasets used along with their imbalance ratio. Clearly, the datasets are extremely imbalanced. Even a null classifier would predict over 99 percent accuracy.

Table 1. Dataset Class Distribution

Dataset	Majority Class	Minority Class	Imbalance Ratio	No. of Features
DS 1:Credit Card Data (Sept'2013)	284315	492	0.0017	28
DS 2:PubChem Bioassay Data (2011)	466438	1454	0.0031	145
DS 3:Yeast Data (1996)	1479	5	0.0033	8
DS 4:Satimage Data (1993)	5809	626	0.1067	36
DS 5:Diabetes Data (2015)	500	268	0.53	8

3.2 Metrics

For evaluating resampling techniques, the following metric is considered.

Imbalance Ratio: For a given two-class dataset, the imbalance ratio is computed as:

$$imbalanceRatio = \frac{\#instances_{minority}}{\#instances_{majority}} \quad (1)$$

The chosen algorithms assume the issue as imbalanced classification problem. We have considered the confusion matrix given in Table 2 for evaluating metrics. However, classical metrics of accuracy and confusion matrix will not be able

Table 2. Confusion Matrix For Evaluating Classification

Predicted	Actual	
	<i>Normal</i>	<i>Fraud</i>
Normal	True Negatives (TN)	False Negatives (FN)
Fraud	False Positives (FP)	True Positives (TP)

to capture the actual fraud identification rate due to skewness in instances of each class. Thus, metrics that balance the detection of both classes have been considered.

Positive Predictive Value or Precision: PPV is a measure of correct positive results among all positive predictions. The positive predictive value (PPV) is defined as:

$$PPV = TP / (TP + FP) \quad (2)$$

Negative Predictive Value: NPV is the proportion of correctly identified negative values (0 here) among all negative predicted ones. The negative predictive value is defined as:

$$NPV = TN / (TN + FN) \quad (3)$$

Specificity: measures ratio of actual negatives that are correctly identified.

$$Specificity = TN / (TN + FP) \quad (4)$$

Sensitivity: computes percentage of actual positives that are correctly identified.

$$Sensitivity = TP / (TP + FP) \quad (5)$$

Balanced Accuracy: is average detection rate obtained on either class.

$$BalancedAccuracy = (Sensitivity + Specificity) / 2 \quad (6)$$

Prevalence: is the term used for how often the condition yes actually occurs.

$$Prevalence = (TP + FN) / (TP + FP + FN + TN) \quad (7)$$

3.3 Machine Learning Algorithms Used

The following popular machine learning algorithms are used for evaluating the classifier results after applying and before applying the resampling techniques on dataset:

K-Nearest Neighbours- KNN classifiers are supervised learning methods that classify new instances based on various similarity measures [7].

Neural Networks(NN)- For fraud detection, the commonly used method are NNs. NN recognize related patterns, predict values based on associative memory of patterns it learnt [8].

Support Vector Machine(SVM)- SVM are used for supervised learning where the data is linearly classified and analyzed [6] [8].

3.4 Results

This study used five datasets to run through different sampling and classification. Results by applying undersampling and oversampling techniques are shown in Table 3 & 4.

Table 5 represents the results of performance evaluation of machine learning algorithms over chosen metrics before apply resampling to the dataset. There are few NaNs in the table where the classifier couldn't detect even a single true positive or true negative for the result set.

The best resampling techniques from each category was selected that's NCL ADASYN and the results of performance evaluation of ML classifier on the dataset obtained by NCL and ADASYN techniques are shown in Table 6-8.

Table 3. Imbalance Ratio of Undersampling Techniques

Techniques Used	Imbalance Ratio				
	DS 1	DS 2	DS 3	DS 4	DS 5
CNN	0.2046	0.2033	0.0568	0.1226	0.53
OSS	0.2340	0.2331	0.1815	0.1137	0.712
NCL	0.2739	0.2739	0.02083	0.2511	0.75
ENN	0.2041	0.2041	0.0033	0.108	0.557
TL	0.2324	0.2324	0.0033	0.1107	0.6802

Table 4. Imbalance Ratio of Oversampling Techniques

Techniques Used	Imbalance Ratio				
	DS 1	DS 2	DS 3	DS 4	DS 5
PDFOS	0.0052	0.0049	0.099	0.215	0.3943
MWMOTE	0.0020	0.0018	0.5986	0.6341	0.696
SMOTE	0.75	0.71	0.754	0.74	0.75
RWO	0.0020	0.0018	0.698	0.715	0.736
ADASYN	0.9999	0.9987	0.9899	0.9799	0.955
WRACOG	0.0017	0.00119	0.819	0.8599	0.866

Figure 1-2 shows the results of balanced accuracy of ML classifiers for NCL and ADASYN re-sampled data.

3.5 Discussion of Results

From the result in Section 3.4, the performance is interpreted as follows:

Testing the effectiveness for Balancing Dataset

In undersampling techniques, NCL showed improved imbalance ratio results for

Table 5. Performance Evaluation of Machine Learning Algorithms Before Resampling

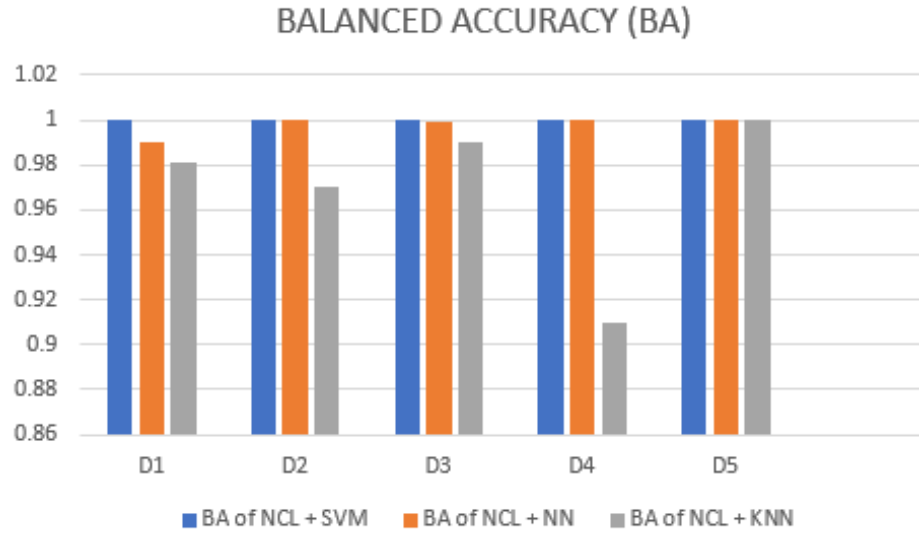
Classifier Metrics		Dataset Used				
Used		DS 1	DS 2	DS 3	DS 4	DS 5
Accuracy	SVM	0.986	0.978	0.985	0.997	0.998
	KNN	0.998	0.985	0.991	0.967	0.997
	NN	0.99825	0.989	0.971	0.986	0.992
PPV	SVM	NaN	NaN	0.015	0.056	NaN
	KNN	NaN	NaN	0.58	0.807	0.851
	NN	0.99	0.891	0.805	0.916	0.956
NPV	SVM	NaN	NaN	0	NaN	0.564
	KNN	0.99	0.891	0.853	0.9020	0.928
	NN	NaN	NaN	0.003	NaN	0.149
Sensitivity	SVM	0.93	0.897	0.896	0.746	0.864
	KNN	0	0	0.210	0.164	0.128
	NN	1	0.976	0.895	0.901	0.945
Specificity	SVM	0.92	0.872	0.81	0.89	0.823
	KNN	1	0.98	0.971	0.996	0.99
	NN	0	0.001	0.002	0	0
Prevalence	SVM	NaN	NaN	0.261	0.5946	0.698
	KNN	0.001	0	0.091	0.566	0.771
	NN	0.99	0.89	0.981	0.996	0.998

Table 6. Performance Evaluation OF Machine Learning Algorithms After Applying NCL

Classifier Values		Dataset Used				
		DS 1	DS 2	DS 3	DS 4	DS 5
Metrics	Classifier Techniques	After Appylyng NCL				
Accuracy	<i>SVM</i>	1	1	1	1	1
	<i>KNN</i>	1	1	1	1	1
	<i>NN</i>	1	1	1	1	1
PPV	<i>SVM</i>	1	1	1	1	1
	<i>KNN</i>	1	1	1	1	1
	<i>NN</i>	1	1	1	1	1
NPV	<i>SVM</i>	1	1	1	1	1
	<i>KNN</i>	1	1	1	1	1
	<i>NN</i>	1	1	1	1	1
Sensitivity	<i>SVM</i>	1	1	1	1	1
	<i>KNN</i>	1	1	1	1	1
	<i>NN</i>	1	1	1	1	1
Specificity	<i>SVM</i>	1	1	1	1	1
	<i>KNN</i>	1	1	1	1	1
	<i>NN</i>	1	1	1	1	1
Prevalence	<i>SVM</i>	0.78	0.79	0.80	0.79	0.69
	<i>KNN</i>	0.69	0.65	0.65	0.79	0.61
	<i>NN</i>	0.81	0.89	0.72	0.78	0.81

Table 7. Performance Evaluation OF Machine Learning Algorithms After Applying ADASYN

Classifier Values		Dataset Used				
		DS 1	DS 2	DS 3	DS 4	DS 5
Metrics	Classifier Techniques	After Appyling ADASYN				
Accuracy	<i>SVM</i>	0.92	0.89	0.90	0.98	0.89
	<i>KNN</i>	0.93	0.81	0.89	0.90	0.87
	<i>NN</i>	0.91	0.83	0.95	0.96	0.87
PPV	<i>SVM</i>	0.91	0.98	0.96	0.99	0.98
	<i>KNN</i>	0.94	0.97	0.98	0.97	0.98
	<i>NN</i>	0.89	0.97	0.98	0.95	0.98
NPV	<i>SVM</i>	0.91	0.97	0.99	0.99	0.82
	<i>KNN</i>	0.94	0.95	0.84	0.80	0.79
	<i>NN</i>	0.89	0.96	0.81	0.89	0.81
Sensitivity	<i>SVM</i>	0.93	0.95	0.98	0.98	0.99
	<i>KNN</i>	0.93	0.94	0.91	0.95	0.99
	<i>NN</i>	0.91	0.95	0.91	0.91	0.99
Specificity	<i>SVM</i>	0.87	0.92	0.77	0.81	0.73
	<i>KNN</i>	0.85	0.89	0.69	0.78	0.79
	<i>NN</i>	0.71	0.90	0.69	0.61	0.80
Prevalence	<i>SVM</i>	0.93	0.81	0.81	0.80	0.79
	<i>KNN</i>	0.89	0.71	0.72	0.63	0.66
	<i>NN</i>	0.91	0.81	0.61	0.95	0.65

**Fig. 1.** Balanced Accuracy After Applying NCL

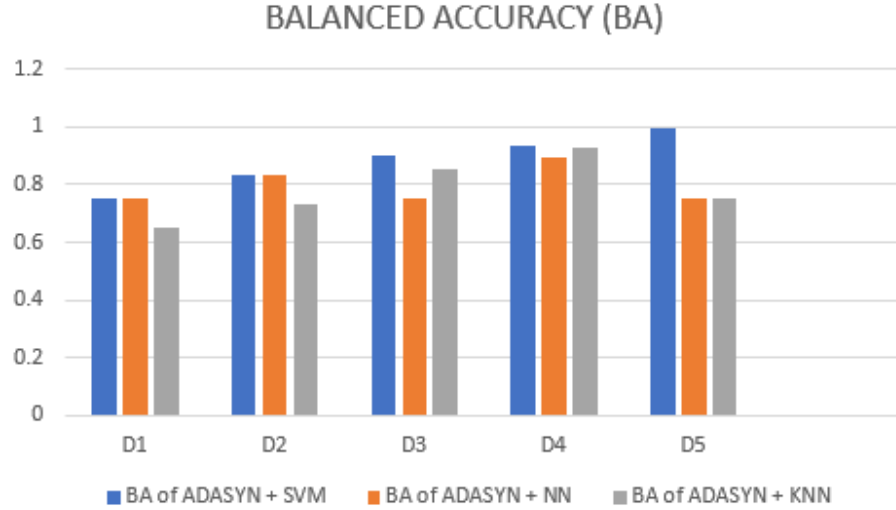


Fig. 2. Balanced Accuracy After Applying ADASYN

all five datasets. In oversampling techniques, ADASYN gave upto 0.99 imbalance ratio. From Table 3-4, out of all the resampling techniques, NCL in undersampling and ADASYN in oversampling showed best overall results. Therefore, we further evaluated these two for checking performance of selected machine learning algorithms.

Performance Evaluation of ML Algorithms

It can be seen from the results in Table 5 that none of the classification mechanisms is best in all respects. Therefore, the results have been scrutinized and interpreted with respect to performance question relevant. There were few NaNs values where the classifier couldnt detect even a single true positive or true negative value. We evaluated these classifiers again after applying NCL and ADASYN resampling techniques to get improved performance.

Performance Evaluation of Classifier After Applying Resampling Techniques

From Table 6-7, we can clearly see that the results have been improved a lot for all ML algorithms. KNN and NN showed similar results for the chosen metrics. But, SVM scored best among all the classifiers used as well as overall also in performance.

Testing overall effectiveness of classification

Balanced Accuracy is used as a measure to detect classification effectiveness in imbalanced datasets. Out of all chosen methods SVM classifier showed overall

perfect results for both NCL under-sampled data and for ADASYN over-sampled data.

The resampling techniques helped to reduce imbalance ratio of the datasets and furthermore removed NaN values for classifiers by improving the skewness of imbalanced datasets for better classification results.

4 Conclusion

In this paper, a number of resampling techniques to improve the imbalance ratio of highly imbalanced datasets has been evaluated. Imbalance data classification is a peculiar classification problem due to very high imbalance in instances of minority and majority class instances in data as examples. Popular Machine Learning algorithms SVM, KNN and NN were evaluated on different metrics. It is concluded that the effect of balancing techniques resulted improved performance of classifier. Undersampling approaches handles the dataset skewness in better ways and hence perform well over all metrics compared to other techniques. It has been observed that ADASYN approach can best improve the imbalance ratio as well as classification results. However, undersampling approaches give better overall performance on all datasets.

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